

Appendix

Loaded words, shifted verdicts: Does the PolBiX x-phemism effect replicate in GPT-4.1 mini and Claude Haiku 4.5?

C Jagadeeswar · NLP Module · Universität Trier · September 2026

A. References

ACADEMIC SOURCES

1. Jakob, C., Harbecke, D., Parschan, P., Wenzel Neves, P., & Schmitt, V. (2025). PolBiX: Detecting LLMs' Political Bias in Fact-Checking through X-phemisms. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. arXiv:2509.15335.
2. Feng, S., Park, C. Y., Liu, Y., & Tsvetkov, Y. (2023). From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023)*.
3. Rozado, D. (2024). The political preferences of LLMs. *PLOS ONE*, 19(7), e0306621. <https://doi.org/10.1371/journal.pone.0306621>
4. Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose Opinions Do Language Models Reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202.
5. De Smet, H. (2005). A corpus of Late Modern English texts. *ICAME Journal*, 29, 69–82.
6. Efron, B., & Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.
7. Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference (SciPy 2010)*.

DATASETS, MODEL DOCUMENTATION AND SOFTWARE

8. Jakob, C. et al. (2025). PolBiX dataset and evaluation scripts (`dataset.csv`, `evaluate.py`, `plot.py`). GitHub: <https://github.com/XplainNLP/PolBiX> (CC BY-NC 4.0).
9. De Smet, H., Flach, S., Diller, H.-J., & Tyrkkö, J. (2015). *The Corpus of Late Modern English Texts, version 3.1 (CLMET3.1)*. <https://fedora.clarin-d.uni-saarland.de/clmet/clmet.html>; plain-text Parquet mirror used: https://huggingface.co/datasets/biglam/clmet_3_1 (CC BY-SA 4.0).
10. OpenAI (2025). Introducing GPT-4.1 in the API (model card / documentation for gpt-4.1-mini). <https://openai.com/index/gpt-4-1/>
11. Anthropic (2025). System Card: Claude Haiku 4.5 (October 2025). <https://www.anthropic.com/claude-haiku-4-5-system-card>

B. Reproducibility

Code, prompts, the 195 CLMET sentence pairs and all raw model outputs:

<https://github.com/USERNAME/polbix-replication>.

1. `run_polbix.py` — prompts both models (temperature 0, 3 runs, max 5 output tokens) with the original German prompts (PolBiX) or their English translations (CLMET); rate-limited, resumable.
2. `build_clmet_pairs.py` — samples candidate sentences from CLMET 3.1 and swaps one noun; the final 195 pairs were then reviewed one by one (AI-assisted) for word sense and grammaticality (`clmet_pairs.csv`).
3. `poster/stats.py`, `poster/make_poster.py` — $\Delta P(\text{true})$, bootstrap CIs and the authors' logistic regression; builds this poster.

Models: `gpt-4.1-mini` (OpenAI API) and `claude-haiku-4-5-20251001` (Anthropic API), queried 28 September 2026.

C. Declaration of Academic Integrity

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(download from the university website, sign, and merge it into this PDF).